Answer Key — Competitive Exam Practice 7: Operations on Processes

GATE (Computer Science) Pattern

Auqib Hamid Lone

Department of Computer Science & Engineering, GCET Ganderbal

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Concepts That Click

Q1 (B). `fork` returns twice: `0` in the child and the child's pid in the parent (`-1` only on failure). Both processes resume at the instruction after the call, not at the start of `main`, so (D) is false.

Q2 (B). `exec` replaces the memory image of the calling process; the pid and the parent relationship are unchanged. A successful `exec` never returns, because the code that called it no longer exists. No new process is created, so (A) and (C) are false; (D) is false because `exec` returns only on failure, with `-1`.

Q3 (B). A zombie has terminated; its resources are released, but its PCB entry (pid and exit status) remains until the parent calls `wait`. Option (A) describes an orphan.

Q4 (C). The child is an orphan; UNIX re-parents it to `init`, which calls `wait` periodically and reaps it. (A) describes cascading termination, which some systems enforce but UNIX does not require.

Q5 63. Six sequential forks give $2^6 = 64$ processes, of which $64 - 1 = 63$ are new children.

Q6 4. First `fork`: P (non-zero) and C1 (zero). C1's `&&` is false, so C1 stops. P evaluates `fork() || fork()`: P sees non-zero from the second `fork` and its `||` is true; the new child C2 sees zero and evaluates the third `fork`, creating C3. Processes P, C1, C2, C3 each print once: 4.

Q7 30. The print is inside the loop. Iterations $i = 0, 1, 2, 3$ run with $2, 4, 8, 16$ processes after their `fork`, each printing once: $2 + 4 + 8 + 16 = 30$.

Q8 8. `"X"` has no newline and output goes to a file, so it remains in the stdio buffer. Each `fork` copies the buffer; three forks give $2^3 = 8$ processes, each of which flushes its own copy of `"X"` at exit: 8 characters.

Q9 18. Parent and child have separate copies of `x`, each starting at 5. The child prints $5 \times 2 = 10$; the parent waits, then prints $5 + 3 = 8$. Sum $= 10 + 8 = 18$.

Q10 44. Only the low 8 bits are delivered: $300 \bmod 256 = 300 - 256 = 44$.